What static transport data determine about a conserving phonon closure

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Non-Fourier interpretations of thermoreflectance experiments on thin films attach kinetic meaning to the relaxation time and the nonlocal length of hydrodynamic heat-flux laws. We ask what static transport data—mode lifetimes, collision invariants and the dc thermal conductivity—determine about the dynamic part of an energy-conserving closure of the linearized phonon Boltzmann equation, and which further operator or measurement information is required. In entropy coordinates the Cattaneo time of the closure is the first memory moment $\tau_{\mathrm{mem}} = -K'(0)/K(0)$ of the current response. Static data bound τ_{mem} from below by $K(0)/|\mathbf{j}|^2$, but in the class of positive-semidefinite collision operators they do not bound it from above: an explicit family keeps all lifetimes, the invariant and $K(0)$ fixed while $\tau_{\mathrm{mem}} \to \infty$. If the collision events are fixed by the harmonic spectrum and only their nonnegative rates are unknown, a convex-hull theorem of Ben-Tal and Teboulle gives $\tau_{\mathrm{mem}} \le M^2 K(0)$ with a sharp, rate-independent constant M, and a per-event lower bound on the rates bounds τ_{mem} without any data. In the relaxation-time approximation the memory moment diverges in the continuum when the infrared exponent α of the acoustic rates satisfies $2\alpha \ge d$, while the conductivity stays finite for $\alpha < d$; for event operators the same divergence follows under an explicit infrared hypothesis, and then $M^2 K(0)$ diverges under Brillouin-zone refinement. For a one-dimensional front-face measurement on a Cattaneo or Guyer–Krumhansl film on a substrate, the depth-scaling gauge of the Fourier problem persists: the film enters only through $(\xi_1, b, \tau_R, \tau_\ell)$, so neither the nonlocal length nor the closure coefficient is separately identifiable. **[PENDING:** abstract sentences on the AlN film-regularized memory-time ranges, FDTR detectability and the rank theorem]

I. INTRODUCTION

Thermoreflectance in the time domain (TDTR) and in the frequency domain (FDTR) is a standard probe of heat transport in thin films and across interfaces [1, 2]. When the heated volume or the modulation period approaches phonon mean free paths or lifetimes, the data depart from the predictions of Fourier's law: apparent conductivities depend on the modulation frequency [3, 4] and on the size and anisotropy of the heating [5], and such departures have been inverted for mean-free-path or relaxation-time spectra [6–8]. A second line of analysis replaces Fourier's law by a constitutive law with a heat-flux relaxation time τ_R—the Cattaneo law [9]—or, in addition, a nonlocal length ℓ—the Guyer–Krumhansl (GK) equation [10–12]—and interprets thermoreflectance signals through these parameters [13–15]. Their values are then read kinetically: τ_R as a resistive (momentum-relaxing) time and ℓ as a length set by normal processes.

Aluminum nitride is a natural test case for this inference. Its thermal conductivity is high—about 321 $\mathrm{W\,m^{-1}K^{-1}}$ at room temperature was measured by TDTR on films grown to 18 and 22.5 μm [16], a value that cannot be transferred to a submicrometer film without a measurement—and first-principles transport workflows for it have been benchmarked against one another [17]; Callaway-type descriptions of its phonon transport have been examined against first-principles rates [18]. A first-principles route from AlN to a thin-film thermoreflectance prediction has three steps: (i) static transport data—mode lifetimes and the bulk conductivity tensor—are computed from harmonic and anharmonic force constants; (ii) an energy-conserving closure of the linearized Boltzmann equation is constructed from them; (iii) the closure enters a model of the film response. The closure coefficients that govern the dynamics, above all the Cattaneo time, are not static quantities. This paper addresses the question: *what do static transport data determine about a conserving dynamic closure, and what further operator or measurement information is needed?*

Several ingredients are established and are used here as attributed background. For the grey Callaway model [19], the conserving closure has grad-div coefficient 1/3 instead of the coefficient 2 of the original GK equation; Guo and Wang give it, including the combined time τ_c, in Eq. (7) of Ref. [20], Sendra *et al.* derive and discuss it [21, 22], and the coefficient 2 has been attributed to an approximate inverse of the normal-process collision matrix [20, 23]; see also Ref. [24]. The eigenvectors of the symmetrized collision operator (relaxons) express the conductivity as a sum over relaxation modes [25], and conductivity and thermal viscosity are carried by relaxons of opposite parity [26]. Reductions of collision dynamics to a few collective variables are exact only under lumpability conditions [27–30] and otherwise generate memory terms of projection type [31, 32], on which a recent statistical theory of heat conduction with spatiotemporal memory kernels is built [33]; descriptions built on diagonal rates alone can fail for hydrodynamic transport [34]. For Fourier films, Krapez and Rigollet showed that thickness, diffusivity and conductivity of a coating are not separately identifiable from front-face

data, whatever the frequencies and noise level [35].

Within this background the paper makes the following contributions. Each is stated with its hypotheses and one of the status labels defined below.

1. In entropy coordinates the Cattaneo time of a conserving closure, defined by matching a single-pole law to the dynamic conductivity at zero frequency, is the first memory moment $\tau_{\mathrm{mem}} = -K'(0)/K(0) = |\mathcal{C}^{+}\mathbf{j}|^2/\mathbf{j}^{\mathsf{T}}\mathcal{C}^{+}\mathbf{j}$ of the current response (Sec. II); in the grey Callaway model it equals τ_R exactly. This is a definition and an elementary check, not a new result.

2. Static data (lifetimes, invariants and the dc response) bound τ_{mem} from below by $\tau_{\mathrm{stat}} = K(0)/|\mathbf{j}|^2$ (Proposition 2) but, in the class of positive-semidefinite operators, not from above (Theorem 1). The mathematics is elementary: it is an instance of the indeterminacy of a Stieltjes function from finitely many of its moments [36, 37]. We have not found an explicit identifiability statement of this kind for phonon kinetics in the literature searched.

3. If the events are fixed by the harmonic spectrum and only their nonnegative rates are unknown, $\tau_{\mathrm{mem}} \le M^2 K(0)$ with a sharp, rate-independent constant M (Theorem 2 and Proposition 3). The inequality is a known consequence of the convex-hull theorem of Ben-Tal and Teboulle for weighted least squares [38]; its application to the memory of kinetic operators is potentially novel within the literature searched. A per-event lower bound on the rates bounds τ_{mem} without any data (Proposition 5).

4. In the relaxation-time approximation (RTA) the memory moment diverges in the continuum when the infrared exponent of the acoustic rates satisfies $2\alpha \ge d$, while the conductivity is finite for $\alpha < d$; for event operators the divergence follows under an explicit hypothesis on the infrared solution (Theorem 3), and then $M^2K(0)$ diverges under mesh refinement. No novelty is claimed for this criterion; an analogous coexistence of a finite conductivity with divergent higher moments, of the free-path distribution, underlies the superdiffusive description of quasiballistic transport in alloys [39, 40].

5. For a one-dimensional front-face measurement on a Cattaneo or GK film on a substrate, the depth-scaling gauge of the Fourier problem persists: the film enters only through $(\xi_1, b, \tau_R, \tau_\ell)$, so neither ℓ nor the closure coefficient is separately identifiable (Proposition 6 and Corollary 2). The Fourier gauge is known [35]; its extension to non-Fourier laws is elementary dimensional analysis and is potentially novel within the literature searched. Whether the four combinations are themselves identifiable is the subject of a theorem stated in Sec. IV: [**PENDING:** Theorem: generic rank four and degenerate set].

6. [**PENDING:** AlN first-principles results: film-regularized memory-time ranges (Sec. V)] [**PENDING:** FDTR detectability of those ranges (Sec. VI)]

The paper does not claim a value of the bulk memory time of AlN, an experimental validation, or any part of the conserving closure itself. The Fourier resonance $\tau_R = \tau_\ell$ used in Sec. IV was named by Both *et al.* [41] and analyzed by Kovács [42].

Status labels. PROVED: a complete proof is given. DERIVED UNDER ASSUMPTIONS: the derivation is complete given hypotheses that are stated but not proved (for instance a property of a specific operator, or a model of the experiment). OPEN: neither proved nor refuted. Numerical results in Secs. V and VI will carry the label NUMERICALLY DEMONSTRATED together with their error estimates. Statements of novelty are restricted to the literature we searched in October 2026: Crossref and OpenAlex (about fifty queries each), arXiv full text, publisher and repository pages, and the forward citations of Refs. [38, 43].

Section II sets up the kinetic framework, the memory moment and the two operator classes. Section III states and proves what static data do and do not determine. Section IV formulates the front-face measurement model and its gauge. Sections V–VII are reserved for the AlN results, the detectability analysis and the conclusions. The notation is collected in Appendix A.

II. LINEARIZED KINETICS IN ENTROPY COORDINATES

A. Modes, entropy coordinates and the collision operator

We work with a finite, discretized description of the phonon gas; continuum limits are taken explicitly in Sec. III E.

(K1) A crystal at uniform temperature $T > 0$ is described on a finite Γ-centered mesh of $N_{\mathbf{k}}$ wave vectors, invariant under $\mathbf{k} \mapsto -\mathbf{k}$; the crystal volume is $\mathcal{V} = N_{\mathbf{k}} V_{\mathrm{cell}}$. The modes $\nu = (\mathbf{k}, s)$, $\nu = 1, \dots, n$, have energies $\varepsilon_\nu = \hbar\omega_\nu > 0$ (zero-frequency modes at Γ are excluded), group velocities $\mathbf{v}_\nu$ and Bose occupations $f^0_\nu = [\exp(\varepsilon_\nu/k_B T) - 1]^{-1}$.

(K2) The occupations obey the linearized Boltzmann equation $\partial_t \delta f_\nu + \mathbf{v}_\nu \cdot \nabla \delta f_\nu = -(\Omega\, \delta f)_\nu$, and the collision operator satisfies detailed balance and the H theorem: with $\mathsf{W} = \mathrm{diag}(w_\nu)$ and $w_\nu = [f^0_\nu(1 + f^0_\nu)]^{1/2}$, the matrix $\Omega\mathsf{W}^2$ is symmetric and positive semidefinite.

In the entropy coordinates

$$y_\nu = \frac{\delta f_\nu}{\mathcal{V}^{1/2} w_\nu} \tag{1}$$

the kinetic equation reads

$$\partial_t \mathbf{y} + \mathsf{V}\cdot\nabla \mathbf{y} = -\mathcal{C}\,\mathbf{y}, \qquad \mathcal{C} = \mathsf{W}^{-1}\Omega\,\mathsf{W} = \mathcal{C}^{\mathsf{T}} \succeq 0, \tag{2}$$

with $\mathsf{V} = \mathrm{diag}(\mathbf{v}_\nu)$. Full linearized collision operators of this kind are computed from first principles [25, 44, 45]. The name of the coordinates refers to the entropy density: expanding the Bose entropy $s_B(f) = k_B[(1+f)\ln(1+f) - f\ln f]$ of each mode to second order gives

$$\delta s = \frac{\delta E}{T} - \frac{k_B}{2}\,|\mathbf{y}|^2 + O(|\mathbf{y}|^3), \qquad \delta E = \mathbf{e}^{\mathsf{T}}\mathbf{y}, \tag{3}$$

where $e_\nu = \mathcal{V}^{-1/2} w_\nu \varepsilon_\nu$, because $s_B'(f_\nu^0) = \varepsilon_\nu/T$ and $s_B''(f_\nu^0) = -k_B/w_\nu^2$. At fixed energy the Euclidean norm of $\mathbf{y}$ measures the entropy deficit, and collisions produce entropy at the rate $k_B\,\mathbf{y}^{\mathsf{T}}\mathcal{C}\mathbf{y} \geq 0$. Energy conservation in collisions gives $\mathcal{C}\mathbf{e} = 0$.

(K3) The energy is the only collision invariant: $\ker\mathcal{C} = \mathrm{span}\{\mathbf{e}\}$.

Hypothesis (K3) excludes exact conservation of crystal momentum (umklapp and other resistive processes are present). If further invariants exist, everything below holds with $\mathcal{H}$ replaced by the orthogonal complement of all invariants. We write $\mathcal{H} = \mathbf{e}^\perp$, $n_{\mathcal{H}} = \dim\mathcal{H} = n-1$ and $\Pi_{\mathcal{H}}$ for the orthogonal projector onto $\mathcal{H}$; $\mathcal{C}$ maps $\mathcal{H}$ into itself and is positive definite there.

The heat-flux density along the Cartesian direction i is $q_i = \mathbf{j}^{(i)\mathsf{T}}\mathbf{y}$ (a flux per unit area, in $\mathrm{W\,m^{-2}}$), with the current vector

$$j_\nu^{(i)} = \mathcal{V}^{-1/2}\, w_\nu\, \varepsilon_\nu\, v_{\nu,i}. \tag{4}$$

Time reversal $\nu = (\mathbf{k}, s) \mapsto \bar\nu = (-\mathbf{k}, s)$ leaves $\mathbf{e}$ invariant and reverses $\mathbf{j}^{(i)}$, so $\mathbf{j}^{(i)} \in \mathcal{H}$. The harmonic quantities

$$\rho c = \frac{|\mathbf{e}|^2}{k_B T^2}, \qquad \Lambda_i = \frac{|\mathbf{j}^{(i)}|^2}{k_B T^2} = \frac{1}{\mathcal{V}}\sum_\nu c_\nu v_{\nu,i}^2, \tag{5}$$

are the volumetric heat capacity and a ballistic weight; $c_\nu = \varepsilon_\nu^2 w_\nu^2/(k_B T^2)$ is the heat capacity of mode ν. The first identity follows from the response $\mathbf{y} = \mathbf{e}\,\delta T/(k_B T^2)$ of the equilibrium to a temperature change δT.

B. Dynamic conductivity and memory kernel

Consider the response to a spatially uniform temperature gradient $\nabla T(t)$ switched on at $t = 0$ (the zero-wave-vector limit). Write the occupations as the local equilibrium $f_\nu^0(T(\mathbf{r},t))$, with the local temperature fixed by the energy density, plus a deviation whose entropy coordinates $\mathbf{y}$ lie in $\mathcal{H}$. The streaming of the local equilibrium, $\mathbf{v}_\nu\cdot\nabla f_\nu^0 = w_\nu^2\varepsilon_\nu\,\mathbf{v}_\nu\cdot\nabla T/(k_B T^2)$, acts as a source with entropy coordinates $\sum_k \mathbf{j}^{(k)}\partial_k T/(k_B T^2) \in \mathcal{H}$; the time derivative of the local equilibrium is proportional to $\mathbf{e}$ and drops out on projection onto $\mathcal{H}$ (its projection onto $\mathbf{e}$ is the energy balance). For a spatially uniform deviation,

$$\partial_t \mathbf{y} = -\mathcal{C}\mathbf{y} - \frac{1}{k_B T^2}\sum_k \mathbf{j}^{(k)}\,\partial_k T(t), \qquad \mathbf{y}(0) = 0. \tag{6}$$

Its solution gives the heat flux in the form of a constitutive law with memory [46],

$$q_i(t) = -\int_0^t \kappa_{ik}(t-t')\,\partial_k T(t')\,dt', \qquad \kappa_{ik}(t) = \frac{\Phi_{ik}(t)}{k_B T^2}, \tag{7}$$

with the current autocorrelation function $\Phi_{ik}(t) = \mathbf{j}^{(i)\mathsf{T}} e^{-\mathcal{C}t}\mathbf{j}^{(k)}$. Under the time convention $e^{+i\omega t}$ of this paper, the Laplace variable is p and the harmonic regime is $p = i\omega$. The transform of Eq. (7) defines the dynamic conductivity

$$\lambda_{ik}(p) = \frac{K_{ik}(p)}{k_B T^2}, \qquad K_{ik}(p) = \mathbf{j}^{(i)\mathsf{T}}(p+\mathcal{C})^{-1}\mathbf{j}^{(k)}. \tag{8}$$

Because the currents lie in $\mathcal{H}$, where $\mathcal{C}$ is invertible, K_{ik} is analytic at $p = 0$ and the dc conductivity is $\lambda_{ik} = K_{ik}(0)/(k_B T^2)$ with $K_{ik}(0) = \mathbf{j}^{(i)\mathsf{T}}\mathcal{C}^+\mathbf{j}^{(k)}$; $\mathcal{C}^+$ is the Moore–Penrose pseudoinverse. A current with a component along an invariant would add a term proportional to $1/p$, and the pseudoinverse must not be used to hide it. From now on we fix one direction and drop the index i: $\mathbf{j}$, $K(p)$, $\Phi(t)$, $\lambda(p)$ and $\lambda = \lambda(0)$.

With the eigenpairs $(\Gamma_k, \mathbf{u}_k)$ of $\mathcal{C}$ restricted to $\mathcal{H}$ ($\Gamma_k > 0$; the relaxons of Ref. [25]),

$$K(p) = \sum_{k=1}^{n-1} \frac{(\mathbf{u}_k^{\mathsf{T}}\mathbf{j})^2}{p+\Gamma_k}. \tag{9}$$

K is the Stieltjes transform of the spectral measure of $\mathcal{C}|_{\mathcal{H}}$ in the state $\mathbf{j}$; its total mass is $|\mathbf{j}|^2 = \Phi(0)$.

C. The memory moment

Definition 1 (Moments)**.** Under (K1)–(K3) and for $\mathbf{j} \in \mathcal{H}\setminus\{0\}$, let $\mathbf{x} = \mathcal{C}^+\mathbf{j}$ (the unique solution in $\mathcal{H}$ of $\mathcal{C}\mathbf{x} = \mathbf{j}$), and define

$$K_0 = K(0) = \mathbf{j}^{\mathsf{T}}\mathbf{x}, \qquad N = |\mathbf{x}|^2,$$
$$\tau_{\mathrm{mem}} = -\frac{K'(0)}{K(0)}, \qquad \tau_{\mathrm{stat}} = \frac{K_0}{|\mathbf{j}|^2}. \tag{10}$$

Lemma 1 (Equivalent forms of the memory moment; PROVED)**.** *Under the hypotheses of Definition 1,*

$$\tau_{\mathrm{mem}} = \frac{N}{K_0} = \frac{|\mathcal{C}^+\mathbf{j}|^2}{\mathbf{j}^{\mathsf{T}}\mathcal{C}^+\mathbf{j}} = \frac{\int_0^\infty t\,\Phi(t)\,dt}{\int_0^\infty \Phi(t)\,dt} = \frac{\sum_k \lambda_k \Gamma_k^{-1}}{\sum_k \lambda_k}, \tag{11}$$

where $\lambda_k = (\mathbf{u}_k^{\mathsf{T}}\mathbf{j})^2/(k_B T^2\,\Gamma_k)$ is the contribution of relaxon k to λ. In particular $\Gamma_{\max}^{-1} \leq \tau_{\mathrm{mem}} \leq \Gamma_{\min}^{-1}$, where $\Gamma_{\min}$ and $\Gamma_{\max}$ are the extreme eigenvalues of $\mathcal{C}|_{\mathcal{H}}$.

Proof. On $\mathcal{H}$, $(p+\mathcal{C})^{-1}$ is analytic at $p=0$ with derivative $-(p+\mathcal{C})^{-2}$; hence $-K'(0)=\mathbf{j}^{\mathsf{T}}(\mathcal{C}^{+})^{2}\mathbf{j}=|\mathcal{C}^{+}\mathbf{j}|^{2}$ because $\mathcal{C}^{+}$ is symmetric. In time, $\int_0^\infty e^{-\mathcal{C}t}dt=\mathcal{C}^{+}$ and $\int_0^\infty t\,e^{-\mathcal{C}t}dt=(\mathcal{C}^{+})^{2}$ on $\mathcal{H}$. The last form follows from Eq. (9): $K_0=\sum_k(\mathbf{u}_k^{\mathsf{T}}\mathbf{j})^2\Gamma_k^{-1}$ and $N=\sum_k(\mathbf{u}_k^{\mathsf{T}}\mathbf{j})^2\Gamma_k^{-2}$. A weighted mean of the Γ_k^{-1} lies between their extremes. □

Thus τ_{mem} is the first moment of the normalized memory kernel $\kappa(t)/\lambda$ and the conductivity-weighted mean relaxon lifetime. In the RTA, where $\mathcal{C}$ is replaced by $\mathrm{diag}(r_\nu)$ (an operator that does not conserve energy), $\tau_{\mathrm{mem}}^{\mathrm{RTA}}=\sum_\nu j_\nu^2 r_\nu^{-2}/\sum_\nu j_\nu^2 r_\nu^{-1}$ and $\tau_{\mathrm{stat}}^{\mathrm{RTA}}=\sum_\nu j_\nu^2 r_\nu^{-1}/\sum_\nu j_\nu^2$.

D. The Cattaneo time of a conserving closure

For spatially uniform fields, Cattaneo's law $\tau_R\partial_t q+q=-\lambda\,\partial_i T$ is Eq. (7) with the exponential kernel $\kappa_C(t)=(\lambda/\tau_R)e^{-t/\tau_R}$, that is, with the single pole $\lambda_C(p)=\lambda/(1+p\tau_R)$ [9, 46].

Definition 2 (Cattaneo time)**.** The Cattaneo time of a conserving operator $\mathcal{C}$ (for a given direction) is the relaxation time of the single pole that matches $\lambda(p)$ and $\lambda'(p)$ at $p=0$. Since $\lambda_C(0)=\lambda$ and $\lambda_C'(0)=-\lambda\tau_R$, it equals τ_{mem}.

The match is exact for all p if and only if Φ is a single exponential, that is, if and only if $\mathbf{j}$ is an eigenvector of $\mathcal{C}$. The matched pole reproduces the slope of $\lambda(p)$ at $p=0$, not the response in a finite frequency band; Theorem 1(v) below shows that the two can be unrelated.

The grey conserving closure illustrates Definition 2. In the Callaway model [19] with a common linear isotropic dispersion $\varepsilon=\hbar v|\mathbf{k}|$ and constant times $\tau_N,\tau_R>0$, the linearized operator in entropy coordinates is

$$\mathcal{C}_{\mathrm{Cal}}=\tau_N^{-1}(I-\Pi_{\mathrm{N}})+\tau_R^{-1}\,\Pi_{\mathcal{H}}, \tag{12}$$

where Π_{N} is the orthogonal projector onto the span of $\mathbf{e}$ and of the three vectors $\mathbf{P}_x$, $\mathbf{P}_y$, $\mathbf{P}_z$, with $P_{i,\nu}=\mathcal{V}^{-1/2}w_\nu\hbar k_{\nu,i}$, which represent crystal momentum: the displaced equilibrium that matches the energy and momentum of a state $\mathbf{y}$ is its orthogonal projection onto that span. We take the mesh symmetric under the cubic point group, so that $\mathbf{e},\mathbf{P}_x,\mathbf{P}_y,\mathbf{P}_z$ are mutually orthogonal.

Proposition 1 (Grey conserving closure; DERIVED UNDER ASSUMPTIONS)**.** *Under the assumptions just stated,* $\mathbf{j}^{(i)}=v^2\mathbf{P}_i$ *and* $\mathcal{C}_{\mathrm{Cal}}\mathbf{j}^{(i)}=\mathbf{j}^{(i)}/\tau_R$*. Hence* $\lambda(p)=\lambda/(1+p\tau_R)$ *exactly, with* $\lambda=\rho c\,v^2\tau_R/3$*, and* $\tau_{\mathrm{mem}}=\tau_{\mathrm{stat}}=\tau_R$*. The mode relaxation rates are* $(\mathcal{C}_{\mathrm{Cal}})_{\nu\nu}=\tau_N^{-1}+\tau_R^{-1}-O(n^{-1})$.

Proof. For $\varepsilon=\hbar v|\mathbf{k}|$, $\varepsilon_\nu v_{\nu,i}=\hbar v^2 k_{\nu,i}$, so $\mathbf{j}^{(i)}=v^2\mathbf{P}_i\in\operatorname{range}\Pi_{\mathrm{N}}\cap\mathcal{H}$, and Eq. (12) gives $\mathcal{C}_{\mathrm{Cal}}\mathbf{j}^{(i)}=\mathbf{j}^{(i)}/\tau_R$. Then $K(p)=|\mathbf{j}|^2/(p+\tau_R^{-1})$, and $\lambda=|\mathbf{j}|^2\tau_R/(k_BT^2)=\Lambda_i\tau_R=\rho c\,v^2\tau_R/3$ by Eq. (5) and cubic symmetry. The diagonal elements of the projectors are $e_\nu^2/|\mathbf{e}|^2+\sum_i P_{i,\nu}^2/|\mathbf{P}_i|^2$ and $e_\nu^2/|\mathbf{e}|^2$, both $O(n^{-1})$. □

Eliminating the deviatoric second moment of the same model under $|p|\tau_N\ll1$ and $v\tau_N/L_{\mathrm{v}}\ll1$, with L_{v} the scale of spatial variation, gives the conserving GK closure [20–22]

$$\tau_R\partial_t\mathbf{q}+\mathbf{q}=-\lambda\nabla T+\ell^2\big[\nabla^2\mathbf{q}+\beta\,\nabla(\nabla\cdot\mathbf{q})\big], \tag{13}$$

with $\beta=1/3$, $\ell^2=v^2\tau_c\tau_R/5$ and $\tau_c=(\tau_N^{-1}+\tau_R^{-1})^{-1}$; the original GK equation has $\beta=2$ and τ_N in place of τ_c [11]. Its relaxation time is the τ_R of Proposition 1, i.e., τ_{mem}. The mode lifetimes, however, are close to τ_c, not to τ_R: lifetimes alone do not give the Cattaneo time even in the grey model, whereas within this two-parameter family the pair (τ_c,λ) fixes (τ_N,τ_R). Section III asks what remains of this when the operator is not restricted to a two-parameter family.

E. Static data and operator classes

Definition 3 (Static data)**.** For $\mathcal{C}$ satisfying (K2)–(K3), the static data are $\mathcal{D}(\mathcal{C})=(\mathbf{r},\ker\mathcal{C},K_0)$, where $\mathbf{r}=\mathrm{diag}\,\mathcal{C}$ are the mode relaxation rates (lifetimes $\tau_\nu=1/r_\nu$). The harmonic vectors $\mathbf{e}$ and $\mathbf{j}$ are fixed. For several directions, K_0 is replaced by the dc tensor.

Under (K3) all $r_\nu>0$: a vanishing diagonal element of a positive-semidefinite matrix forces the corresponding row to vanish, which would put a unit vector in $\ker\mathcal{C}=\mathrm{span}\{\mathbf{e}\}$, whereas $\mathbf{e}$ has no zero component. The rates r_ν include the self-coupling of repeated slots in an event; they are not, in general, the self-energy linewidths of RTA codes (see Lemma 2).

Definition 4 (Operator classes)**.** (P) The positive-semidefinite class is $\mathfrak{P}(\mathbf{e})=\{\mathcal{C}\in\mathbb{R}^{n\times n}:\mathcal{C}=\mathcal{C}^{\mathsf{T}}\succeq0,\ \ker\mathcal{C}=\mathrm{span}\{\mathbf{e}\}\}$. (E) Let $Z=(\boldsymbol{\zeta}_\gamma)_{\gamma\in\mathcal{E}}$ be a fixed finite family of vectors in $\mathcal{H}$ that spans $\mathcal{H}$. For rates $g\in[0,\infty)^{\mathcal{E}}$ put

$$\mathcal{C}(g)=\sum_{\gamma\in\mathcal{E}}g_\gamma\,\boldsymbol{\zeta}_\gamma\boldsymbol{\zeta}_\gamma^{\mathsf{T}}. \tag{14}$$

The rates are admissible if $\ker\mathcal{C}(g)=\mathrm{span}\{\mathbf{e}\}$, equivalently if $\{\boldsymbol{\zeta}_\gamma:g_\gamma>0\}$ spans $\mathcal{H}$. The fixed event cone is $\mathfrak{E}(Z)=\{\mathcal{C}(g):g\text{ admissible}\}\subset\mathfrak{P}(\mathbf{e})$.

Three-phonon kinetics has the form (14) [45]. For an event $\gamma:\ \mathsf{p}\to\mathsf{a}+\mathsf{b}$ allowed by crystal-momentum conservation (normal or umklapp) and by energy conservation on the mesh, let $\mathbf{s}_\gamma=-\boldsymbol{\delta}_{\mathsf{p}}+\boldsymbol{\delta}_{\mathsf{a}}+\boldsymbol{\delta}_{\mathsf{b}}$ be its stoichiometric vector ($-\boldsymbol{\delta}_{\mathsf{p}}+2\boldsymbol{\delta}_{\mathsf{a}}$ for a repeated daughter). Linearizing the net rate $G_\gamma[f_{\mathsf{p}}(1+f_{\mathsf{a}})(1+f_{\mathsf{b}})-(1+f_{\mathsf{p}})f_{\mathsf{a}}f_{\mathsf{b}}]$ about f^0 and using detailed balance gives $-G_\gamma P_\gamma\,\mathbf{s}_\gamma^{\mathsf{T}}\mathsf{W}^{-2}\delta\mathbf{f}$, with $P_\gamma=f^0_{\mathsf{p}}(1+f^0_{\mathsf{a}})(1+f^0_{\mathsf{b}})$, so that Eq. (2) holds with

$$\boldsymbol{\zeta}_\gamma=\mathsf{W}^{-1}\mathbf{s}_\gamma,\qquad g_\gamma=G_\gamma P_\gamma\geq0. \tag{15}$$

Energy conservation $\mathbf{s}_\gamma^{\mathsf{T}}\boldsymbol{\varepsilon} = 0$ gives $\boldsymbol{\zeta}_\gamma \in \mathcal{H}$. The event set, hence Z, is fixed by the harmonic spectrum; the rates g_γ are fixed by the anharmonic amplitudes, which are the unknowns of the inverse problem considered here. On a finite mesh, exact resonance is exceptional; with a broadened resonance condition the accepted events must be given an energy-conserving stoichiometry, which is a declared approximation (Supplemental Material, Sec. S1). The class statements below concern any fixed finite Z. For given data $(\mathbf{r}^\star, K^\star)$ the feasible set is

$$\mathcal{F}(\mathbf{r}^\star, K^\star) = \{g \text{ admissible} : \operatorname{diag}\mathcal{C}(g) = \mathbf{r}^\star, \; K_0(g) = K^\star\}. \tag{16}$$

Compressing the kinetics onto fewer variables is a separate question. Block observables $\mathbf{N} = \mathsf{B}\mathbf{y}$ (for instance the populations of degenerate blocks) obey a closed equation $\dot{\mathbf{N}} = -\mathsf{G}\mathbf{N}$ for every initial state if and only if $\mathsf{B}\mathcal{C} = \mathsf{G}\mathsf{B}$, the exact-lumpability condition [27–30]; otherwise the reduced dynamics contains memory terms of projection type [31, 32]. We use this known criterion only as background.

III. STATIC DATA DO NOT DETERMINE THE MEMORY TIME

Fix the harmonic vectors $\mathbf{e}$ and $\mathbf{j} \in \mathcal{H}\backslash\{0\}$ and an operator class $\mathfrak{X} \in \{\mathfrak{P}(\mathbf{e}), \mathfrak{E}(Z)\}$. The question of this section is the size of the set $\{\tau_{\mathrm{mem}}(\mathcal{C}) : \mathcal{C} \in \mathfrak{X},\ \mathcal{D}(\mathcal{C}) = \mathcal{D}^\star\}$ for given static data $\mathcal{D}^\star$. In Example 1 and Theorem 1 rates are measured in units of a common rate and times in its inverse.

A. The positive-semidefinite class

Example 1 (Same data, different memory; PROVED)**.** Let $n = 4$, $\mathbf{e} = (1,1,1,1)^{\mathsf{T}}/2$ and $\mathbf{j} = (1,0,0,-1)^{\mathsf{T}}$. Let $\mathcal{C}_A = \frac{4}{3}I - \frac{1}{3}\mathbf{1}\mathbf{1}^{\mathsf{T}}$ and $\mathcal{C}_B = I - \frac{1}{2}\mathsf{A}_4$, where $\mathbf{1} = (1,1,1,1)^{\mathsf{T}}$ and A_4 is the adjacency matrix of the cycle 1–2–3–4–1. Both operators have unit diagonal and kernel $\operatorname{span}\{\mathbf{e}\}$, and

$$K_A(p) = \frac{2}{p + 4/3}, \qquad K_B(p) = \frac{1}{p+1} + \frac{1}{p+2}. \tag{17}$$

Hence $K_A(0) = K_B(0) = 3/2$, whereas $\tau_{\mathrm{mem}} = 3/4$ for $\mathcal{C}_A$ and $5/6$ for $\mathcal{C}_B$.

Both operators are weighted graph Laplacians on the four modes, so both lie in the single event cone generated by the six vectors $\boldsymbol{\delta}_\mu - \boldsymbol{\delta}_\nu$. Static data therefore determine neither the operator nor τ_{mem} even inside a fixed cone; Theorem 2 below shows that they bound τ_{mem} there. In the positive-semidefinite class no bound exists.

Theorem 1 (Unbounded memory at fixed static data; PROVED)**.** *Let $n = 8$ and label the modes by $i \in \{0,\dots,7\}$. Let H be the symmetric orthogonal Walsh matrix $\mathsf{H}_{ik} = 8^{-1/2}(-1)^{\langle i,k\rangle}$, where $\langle i,k\rangle$ counts the binary digits equal to one in both i and k, and let $\mathbf{h}_k$ be its columns. For $0 < \eta < 1$ let $\mathsf{T}(\eta)$ be the symmetric matrix whose only nonzero entries are*

$$\begin{aligned} \operatorname{diag}\mathsf{T} &= (0, \eta^2, 2, 2-\eta^2, 1, 1, 1, 1), \\ \mathsf{T}_{12} = \mathsf{T}_{21} &= \eta, \qquad \mathsf{T}_{47} = \mathsf{T}_{74} = -\eta, \end{aligned} \tag{18}$$

and put $\mathcal{C}_\eta = \mathsf{H}\,\mathsf{T}(\eta)\,\mathsf{H}^{\mathsf{T}}$, $\mathbf{e} = \mathbf{h}_0$ and $\mathbf{j} = \mathbf{h}_2$. Then:

(i) $\mathcal{C}_\eta \in \mathfrak{P}(\mathbf{e})$;

(ii) $\operatorname{diag}\mathcal{C}_\eta = (1,\dots,1)$;

(iii) $K_\eta(p) = (p+\eta^2)/[p^2+(2+\eta^2)p+\eta^2]$, hence $K_\eta(0) = 1$ and $\tau_{\mathrm{mem}}(\mathcal{C}_\eta) = 1 + \eta^{-2}$;

(iv) the fixed-point-free involution $i \mapsto i \oplus 3$ (bitwise exclusive or) commutes with $\mathcal{C}_\eta$, leaves $\mathbf{e}$ invariant and reverses $\mathbf{j}$;

(v) for $|\omega| \geq \omega_{\min} > 0$, $\left|K_\eta(i\omega)\,(i\omega + 2) - 1\right| \leq \eta^2/(2\omega_{\min})$.

Consequently, in $\mathfrak{P}(\mathbf{e})$ the static data $(\mathbf{r}, \ker\mathcal{C}, K_0) = \big((1,\dots,1), \operatorname{span}\{\mathbf{e}\}, 1\big)$ are compatible with every value $\tau_{\mathrm{mem}} \in (2,\infty)$.

Proof. (i) H is the character table of $(\mathbb{Z}_2)^3$ normalized to be orthogonal; it is symmetric. $\mathsf{T}(\eta)$ is block diagonal on the index sets $\{0\}, \{1,2\}, \{3\}, \{4,7\}, \{5\}, \{6\}$. The block on $\{1,2\}$ has determinant $\eta^2 > 0$ and positive trace; the block on $\{4,7\}$ has determinant $1-\eta^2 > 0$; the remaining diagonal entries are 0, $2-\eta^2 > 0$, 1, 1. Hence $\mathsf{T} \succeq 0$ with kernel $\operatorname{span}\{\boldsymbol{\delta}_0\}$, and $\mathcal{C}_\eta \succeq 0$ with kernel $\operatorname{span}\{\mathsf{H}\boldsymbol{\delta}_0\} = \operatorname{span}\{\mathbf{e}\}$. (ii) Since $\mathsf{H}_{ik}^2 = 1/8$ and $\operatorname{tr}\mathsf{T} = 8$, $(\mathcal{C}_\eta)_{ii} = 1 + 2\eta(\mathsf{H}_{i1}\mathsf{H}_{i2} - \mathsf{H}_{i4}\mathsf{H}_{i7})$. Parity of $\langle i,\cdot\rangle$ is additive under exclusive or, so $\mathsf{H}_{i1}\mathsf{H}_{i2} = \frac{1}{8}(-1)^{\langle i,3\rangle} = \mathsf{H}_{i4}\mathsf{H}_{i7}$, and the bracket vanishes. (iii) $(p + \mathcal{C}_\eta)^{-1} = \mathsf{H}(p + \mathsf{T})^{-1}\mathsf{H}^{\mathsf{T}}$, so $K_\eta(p)$ is the $(2,2)$ entry of the inverse of $p + \mathsf{T}$ restricted to $\{1,2\}$, which is $(p+\eta^2)/[(p+\eta^2)(p+2)-\eta^2]$. At $p = 0$ this is 1; differentiating, $-K_\eta'(0) = 1+\eta^{-2}$. Equivalently, $\mathcal{C}_\eta(-\mathbf{h}_1/\eta + \mathbf{h}_2) = \mathbf{j}$, so $\mathbf{x} = -\mathbf{h}_1/\eta + \mathbf{h}_2$, $K_0 = 1$ and $N = 1 + \eta^{-2}$. (iv) Let P be the permutation $i \mapsto i \oplus 3$. Then $\mathsf{P}\mathbf{h}_k = (-1)^{\langle 3,k\rangle}\mathbf{h}_k$, so P commutes with $\mathcal{C}_\eta$ if and only if T couples only indices of equal sign $(-1)^{\langle 3,k\rangle}$. This holds: the signs are -1 for $k = 1,2$ and $+1$ for $k = 4,7$. The sign is $+1$ for $k = 0$ and -1 for $k = 2$. (v) $K_\eta(p)(p+2)-1 = \eta^2/D(p)$ with $D(p) = p^2+(2+\eta^2)p+\eta^2$, and $|D(i\omega)| \geq |\operatorname{Im} D(i\omega)| = (2+\eta^2)|\omega| \geq 2|\omega|$. For the consequence, take $\eta = (\tau_{\mathrm{mem}} - 1)^{-1/2}$. □

Remark 1 (Mechanism). D has a root $-\eta^2/2 + O(\eta^4)$ whose residue in K_η is $\eta^2/4 + O(\eta^4)$: a relaxation mode of vanishing current weight and diverging lifetime carries about half of K_0 and all of the divergence of N. Part (v) shows that at frequencies $|\omega| \geq \omega_{\min}$ the response is within a relative $\eta^2/(2\omega_{\min})$ of the single fast pole $1/(p+2)$. A band-limited measurement does not see the divergence of τ_{mem}; the slope at $p = 0$ is not a property of the response in a finite band.

Remark 2 (Spectral reading and attribution). By Eq. (9), $|\mathbf{j}|^2$, K_0 and N are the moments of orders 0, -1 and -2 of the spectral measure of $\mathcal{C}|_{\mathcal{H}}$ in the state $\mathbf{j}$. The static data and the harmonic spectrum fix the first two, and τ_{mem} requires the third. Without further structure no bound follows: this is the elementary indeterminacy of a Stieltjes function from finitely many of its moments [36, 37]; the same representation underlies classical bounds on effective transport coefficients of composites [47]. A spectral gap $\Gamma_{\min}$ gives $\tau_{\mathrm{mem}} \le \Gamma_{\min}^{-1}$ (Lemma 1). The content of Theorem 1 is that prescribing the diagonal and the kernel does not supply such structure; after a diagonal scaling, the positive-semidefinite matrices with prescribed diagonal form an affine image of the elliptope [48]. The physical expectation is anticipated by the parity structure of relaxons [26] and by numerical failures of rate-only descriptions [34]; we have not found the identifiability statement itself for phonon kinetics in the literature searched.

Remark 3 (The family is not microscopic). No fixed finite event cone contains $\mathcal{C}_\eta$ for all small η. Let $\mathbf{u} = \mathbf{h}_1$; then $\mathbf{u}^{\mathsf{T}}\mathcal{C}_\eta\mathbf{j} = \eta$ and $\mathbf{u}^{\mathsf{T}}\mathcal{C}_\eta\mathbf{u} = \eta^2$. For $\mathcal{C} = \sum_\gamma g_\gamma \boldsymbol{\zeta}_\gamma\boldsymbol{\zeta}_\gamma^{\mathsf{T}}$ with $g \ge 0$,

$$|\mathbf{u}^{\mathsf{T}}\mathcal{C}\mathbf{j}| \le \sum_\gamma g_\gamma |\mathbf{u}^{\mathsf{T}}\boldsymbol{\zeta}_\gamma|\,|\mathbf{j}^{\mathsf{T}}\boldsymbol{\zeta}_\gamma| \le R\,\mathbf{u}^{\mathsf{T}}\mathcal{C}\mathbf{u}, \tag{19}$$

with $R = \max\{|\mathbf{j}^{\mathsf{T}}\boldsymbol{\zeta}_\gamma|/|\mathbf{u}^{\mathsf{T}}\boldsymbol{\zeta}_\gamma| : \mathbf{u}^{\mathsf{T}}\boldsymbol{\zeta}_\gamma \ne 0\}$ (events with $\mathbf{u}^{\mathsf{T}}\boldsymbol{\zeta}_\gamma = 0$ do not contribute). Membership for all small η would require $\eta \le R\eta^2$, which fails for $\eta < 1/R$; if $R = 0$ or no event has $\mathbf{u}^{\mathsf{T}}\boldsymbol{\zeta}_\gamma \ne 0$, the cross term vanishes, which contradicts $\eta > 0$.

B. What static data do determine

Proposition 2 (Lower bound from static data; PROVED). *For every $\mathcal{C} \in \mathfrak{P}(\mathbf{e})$ and $\mathbf{j} \in \mathcal{H} \setminus \{0\}$,*

$$\tau_{\mathrm{mem}} \;\ge\; \frac{|\mathbf{x}|}{|\mathbf{j}|} \;\ge\; \tau_{\mathrm{stat}} = \frac{K_0}{|\mathbf{j}|^2}, \tag{20}$$

and $\tau_{\mathrm{mem}} = \tau_{\mathrm{stat}}$ if and only if $\mathbf{j}$ is an eigenvector of $\mathcal{C}$.

Proof. By the Cauchy–Schwarz inequality $K_0 = \mathbf{j}^{\mathsf{T}}\mathbf{x} \le |\mathbf{j}|\,|\mathbf{x}|$, so $|\mathbf{x}|/K_0 \ge 1/|\mathbf{j}|$ and $\tau_{\mathrm{mem}} = |\mathbf{x}|\,(|\mathbf{x}|/K_0) \ge |\mathbf{x}|/|\mathbf{j}| \ge K_0/|\mathbf{j}|^2$. Equality throughout holds if and only if $\mathbf{x} \parallel \mathbf{j}$, that is, $\mathcal{C}\mathbf{j} \parallel \mathbf{j}$. □

By Eq. (5), $\tau_{\mathrm{stat}} = \lambda/\Lambda_i$ is the time of the elementary kinetic estimate $\lambda = \Lambda_i\tau$; it is computable from the dc conductivity and the harmonic spectrum. In the RTA it is the $c_\nu v_{\nu,i}^2$-weighted mean lifetime, and Proposition 2 reduces to $\tau_{\mathrm{mem}}^{\mathrm{RTA}} \ge \tau_{\mathrm{stat}}^{\mathrm{RTA}}$. Equality holds in the grey Callaway model (Proposition 1). Static data thus fix a lower end of the range of τ_{mem}; whether that end is approached in a given event geometry is a numerical question [**PENDING:** cross-reference: minima of $\tau_{\mathrm{mem}}/\tau_{\mathrm{stat}}$ found in the Debye and AlN event geometries, Sec. V].

The static data are not independent of each other. For an event operator, the dc response is bounded below by the RTA response built from the same diagonal.

Lemma 2 (Consistency of lifetimes and dc response; PROVED). *Let g be admissible, $\mathbf{r} = \operatorname{diag}\mathcal{C}(g)$ and $K_{\mathrm{RTA}} = \sum_\nu j_\nu^2/r_\nu$, and let $\varsigma_{\max}$ be the largest number of distinct modes touched by an active event (3 for $\mathsf{p} \to \mathsf{a} + \mathsf{b}$ with $\mathsf{a} \ne \mathsf{b}$, 2 for $\mathsf{p} \to 2\mathsf{a}$, 4 for four-phonon events). Then $K_0 \ge K_{\mathrm{RTA}}/\varsigma_{\max}$, and the constant is sharp.*

Proof. For $\mathbf{j} \in \operatorname{range}\mathcal{C}$, $K_0 = \max_{\mathbf{y}}(2\mathbf{j}^{\mathsf{T}}\mathbf{y} - \mathbf{y}^{\mathsf{T}}\mathcal{C}\mathbf{y})$, the variational principle of the linearized Boltzmann equation [49]. For a vector $\boldsymbol{\zeta}$ with $|\operatorname{supp}\boldsymbol{\zeta}| = s$, the Cauchy–Schwarz inequality gives $(\boldsymbol{\zeta}^{\mathsf{T}}\mathbf{y})^2 \le s\sum_\nu \zeta_\nu^2 y_\nu^2$, that is $\boldsymbol{\zeta}\boldsymbol{\zeta}^{\mathsf{T}} \preceq s\operatorname{diag}(\boldsymbol{\zeta}\circ\boldsymbol{\zeta})$; summing with the rates, $\mathcal{C} \preceq \varsigma_{\max}\operatorname{diag}\mathcal{C}$. With $\mathbf{y} = \operatorname{diag}(\mathbf{r})^{-1}\mathbf{j}/\varsigma_{\max}$, $K_0 \ge 2K_{\mathrm{RTA}}/\varsigma_{\max} - K_{\mathrm{RTA}}/\varsigma_{\max}$. Exact networks attain the ratio 1/3 (eleven modes, distinct daughters) and 1/2 (a chain of repeated-daughter events); see the Supplemental Material, Sec. S2. □

The lemma refers to $\operatorname{diag}\mathcal{C}$, in which the self-coupling of a repeated daughter is counted in full. A self-energy linewidth that counts it with half weight is a different quantity, and the inequality can fail for it (an exact counterexample is given in the Supplemental Material, Sec. S2).

C. Fixed event cone

Definition 5 (Basic solutions and the constant M). For $I \subset \mathcal{E}$ with $|I| = n_{\mathcal{H}} - 1$, consider the linear system

$$\boldsymbol{\zeta}_\gamma^{\mathsf{T}}\mathbf{y} = 0 \quad (\gamma \in I), \qquad \mathbf{y} \in \mathcal{H}, \qquad \mathbf{j}^{\mathsf{T}}\mathbf{y} = 1. \tag{21}$$

It has a unique solution $\mathbf{y}_I$ if and only if $\{\boldsymbol{\zeta}_\gamma\}_{\gamma\in I}\cup\{\mathbf{j}\}$ is a basis of $\mathcal{H}$; such I are called bases, and $\mathfrak{B}$ denotes their set, which is nonempty because Z spans $\mathcal{H}$ and $\mathbf{j} \ne 0$. Put $M = M(Z,\mathbf{j}) = \max_{I\in\mathfrak{B}}|\mathbf{y}_I|$.

M does not depend on the rates. It depends on Z only through the hyperplanes $\operatorname{span}\{\boldsymbol{\zeta}_\gamma\}_{\gamma\in I}$, and $M(Z,s\mathbf{j}) = M(Z,\mathbf{j})/|s|$, so that M^2K_0 and $M^2|\mathbf{j}|^2$ do not depend on the normalization of $\mathbf{j}$.

Theorem 2 (Cone bound; PROVED). *Let Z span $\mathcal{H}$, $\mathbf{j} \in \mathcal{H}\setminus\{0\}$ and g admissible. Fix an orthonormal basis of $\mathcal{H}$; for $I \in \mathfrak{B}$ define $\boldsymbol{\psi}_I \in \mathcal{H}$ by $\boldsymbol{\psi}_I^{\mathsf{T}}\mathbf{y} = \det[\boldsymbol{\zeta}_I, \mathbf{y}]$ for all $\mathbf{y} \in \mathcal{H}$, the determinant having the columns $\boldsymbol{\zeta}_\gamma$ ($\gamma \in I$, in a fixed order) followed by $\mathbf{y}$, and put $g_I = \prod_{\gamma\in I} g_\gamma$. Then*

$$\frac{\mathbf{x}}{K_0} = \sum_{I\in\mathfrak{B}} \varpi_I\,\mathbf{y}_I, \qquad \varpi_I = \frac{g_I\,(\boldsymbol{\psi}_I^{\mathsf{T}}\mathbf{j})^2}{\sum_{J\in\mathfrak{B}} g_J\,(\boldsymbol{\psi}_J^{\mathsf{T}}\mathbf{j})^2}, \tag{22}$$

a convex combination over the active bases ($g_I > 0$). Consequently

$$|\mathcal{C}(g)^+\mathbf{j}| \le M\,K_0(g), \qquad \tau_{\mathrm{mem}}(g) \le M^2K_0(g). \tag{23}$$

Proof. Identify $\mathcal{H}$ with $\mathbb{R}^{n_\mathcal{H}}$ through the orthonormal basis and let $A = \mathcal{C}(g)|_\mathcal{H} = Z\,\mathrm{diag}(g)\,Z^\mathsf{T}$, where Z also denotes the matrix with columns $\boldsymbol{\zeta}_\gamma$; A is positive definite. For $\mathbf{y}, \mathbf{z} \in \mathcal{H}$ the matrix determinant lemma gives $\det(A + \mathbf{y}\mathbf{z}^\mathsf{T}) = \det A + \mathbf{z}^\mathsf{T}\mathrm{adj}(A)\,\mathbf{y}$. Writing $A + \mathbf{y}\mathbf{z}^\mathsf{T} = [Z, \mathbf{y}]\,\mathrm{diag}(g, 1)\,[Z, \mathbf{z}]^\mathsf{T}$ and expanding by the Cauchy–Binet formula over the $n_\mathcal{H}$-element subsets of columns, the subsets without the last column give $\det A$ and those containing it give $\sum_{|I|=n_\mathcal{H}-1} g_I \det[\boldsymbol{\zeta}_I, \mathbf{y}] \det[\boldsymbol{\zeta}_I, \mathbf{z}]$. Hence

$$\mathrm{adj}(A) = \sum_{|I|=n_\mathcal{H}-1} g_I\, \boldsymbol{\psi}_I \boldsymbol{\psi}_I^\mathsf{T}, \tag{24}$$

where dependent subsets give $\boldsymbol{\psi}_I = 0$. With $A^{-1} = \mathrm{adj}(A)/\det A$, $\mathbf{x} = \sum_I g_I \boldsymbol{\psi}_I(\boldsymbol{\psi}_I^\mathsf{T}\mathbf{j})/\det A$ and $K_0 = \sum_I g_I(\boldsymbol{\psi}_I^\mathsf{T}\mathbf{j})^2/\det A > 0$. Now $\boldsymbol{\psi}_I^\mathsf{T}\mathbf{j} = \det[\boldsymbol{\zeta}_I, \mathbf{j}] \neq 0$ exactly when $I \in \mathfrak{B}$, and then $\boldsymbol{\psi}_I/(\boldsymbol{\psi}_I^\mathsf{T}\mathbf{j})$ satisfies Eq. (21), because $\boldsymbol{\psi}_I^\mathsf{T}\boldsymbol{\zeta}_\gamma$ is a determinant with a repeated column for $\gamma \in I$; it is therefore $\mathbf{y}_I$. Dividing $\mathbf{x}$ by K_0 gives Eq. (22). Finally $|\mathbf{x}|/K_0 \le \sum_I \varpi_I |\mathbf{y}_I| \le M$ and $\tau_\mathrm{mem} = (|\mathbf{x}|/K_0)^2 K_0$. □

Remark 4 (Relation to weighted least squares). By Lagrange's rule, $\mathbf{x}/K_0$ minimizes $\sum_\gamma g_\gamma(\boldsymbol{\zeta}_\gamma^\mathsf{T}\mathbf{y})^2$ over $\mathbf{y} \in \mathcal{H}$ subject to $\mathbf{j}^\mathsf{T}\mathbf{y} = 1$: it is the solution of a weighted least-squares problem with one exact constraint. Ben-Tal and Teboulle proved, by the same Cauchy–Binet argument, that the solution of a weighted least-squares problem lies in the convex hull of the solutions of its square subsystems for every positive weight vector [38, Theorem 2.1 and Corollary 2.1] (see also Ref. [50]). Theorem 2 is that statement for the constrained problem, which is the limit of infinite weight on the row $\mathbf{j}^\mathsf{T}\mathbf{y} = 1$. Weight-independent bounds on scaled projections and pseudoinverses are classical [43, 51–53] and govern the complexity of interior-point methods [54]; approximating a closely related weight-uniform condition measure is NP-hard in general [55], so for large event sets M must be bounded by exploiting structure. We claim no novelty for the inequality; its use to bound the memory moment of a kinetic operator by its dc response is potentially novel within the literature searched.

The bound extends to rates that are not admissible but satisfy $\mathbf{j} \in \mathrm{range}\,\mathcal{C}(g)$. Put $g_\theta = g + \theta\mathbf{1}$ ($\theta > 0$), which is admissible, and $E = \mathcal{C}(\mathbf{1})$. Split $\mathcal{H} = \mathcal{R} \oplus \mathcal{Q}$ with $\mathcal{R} = \mathrm{range}\,\mathcal{C}(g)$ and $\mathcal{Q} = \ker\mathcal{C}(g) \cap \mathcal{H}$, and write $\mathbf{x}_\theta = \mathcal{C}(g_\theta)^+\mathbf{j} = \mathbf{w}_\theta + \mathbf{u}_\theta$ accordingly. The $\mathcal{Q}$ component of $\mathcal{C}(g_\theta)\mathbf{x}_\theta = \mathbf{j}$ gives $\mathbf{u}_\theta = -(\Pi_\mathcal{Q} E \Pi_\mathcal{Q})^{-1}\Pi_\mathcal{Q} E\,\mathbf{w}_\theta$, and the $\mathcal{R}$ component shows $\mathbf{w}_\theta \to \mathbf{w} = \mathcal{C}(g)^+\mathbf{j}$ as $\theta \to 0$; hence $\mathbf{u}_\theta \to \mathbf{u}_\infty$ and $K_0(g_\theta) = \mathbf{j}^\mathsf{T}\mathbf{w}_\theta \to K_0(g)$. Since $\mathbf{w} \perp \mathbf{u}_\infty$, $|\mathbf{w}| \le |\mathbf{w} + \mathbf{u}_\infty| \le M K_0(g)$ by Theorem 2 and continuity.

Proposition 3 (Sharpness without lifetime constraints; PROVED)**.** *For every $K^\star > 0$,*

$$\sup\{\tau_\mathrm{mem}(g) :\ g \text{ admissible},\ K_0(g) = K^\star\} = M^2 K^\star. \tag{25}$$

The supremum is approached by rates concentrated on a basis I^* with $|\mathbf{y}_{I^*}| = M$ and is in general not attained; along the approaching sequence the rates on I^* diverge, and so do the relaxation rates of the modes touched by I^*.

Proof. For $0 < \epsilon < 1$ let $g_\gamma^\epsilon = 1$ on I^* and ϵ elsewhere; g^ϵ is admissible. For $I \in \mathfrak{B}$, $g_I^\epsilon = \epsilon^{|I \setminus I^*|}$ with $|I \setminus I^*| \ge 1$ for $I \neq I^*$, so $\varpi_{I^*} \to 1$ in Eq. (22) and $|\mathbf{x}|/K_0 \to M$ as $\epsilon \to 0$. Since $\mathcal{C}(sg) = s\mathcal{C}(g)$, the ratio $|\mathbf{x}|/K_0$ is invariant under $g \mapsto sg$ while K_0 scales as $1/s$; rescaling g^ϵ to $K_0 = K^\star$ gives $\tau_\mathrm{mem} = (|\mathbf{x}|/K_0)^2 K^\star \to M^2 K^\star$, and Theorem 2 gives the reverse inequality. Every $n_\mathcal{H}$-element subset of events contains an event outside I^*, and at least one such subset is a basis of $\mathcal{H}$, so $\det A = \Theta(\epsilon)$ while $\sum_I g_I^\epsilon(\boldsymbol{\psi}_I^\mathsf{T}\mathbf{j})^2 \to (\boldsymbol{\psi}_{I^*}^\mathsf{T}\mathbf{j})^2 > 0$; thus $K_0(g^\epsilon) \sim c/\epsilon$ and the rescaling factor $K_0(g^\epsilon)/K^\star$ diverges. □

Remark 5 (With lifetimes fixed). Since $\mathcal{F}(\mathbf{r}^\star, K^\star)$ is contained in the set of Eq. (25), $\sup_\mathcal{F} \tau_\mathrm{mem} \le M^2 K^\star$. Equality is possible: if $\mathbf{j} = \boldsymbol{\zeta}_\beta$ is parallel to an event vector, no subset containing β is a basis, M is attained by some $J^* \not\ni \beta$, and every g supported on the basis $J^* \cup \{\beta\}$ of $\mathcal{H}$ gives $\mathbf{x}/K_0 = \mathbf{y}_{J^*}$ exactly; with $\mathbf{r}^\star = \mathrm{diag}\,\mathcal{C}(g)$ and $K^\star = K_0(g)$ the lifetime-constrained supremum is then attained (an exact example with seven modes and ten events is given in the Supplemental Material, Sec. S3). Physical currents (4) have full support and are never parallel to a three-mode event vector; for them it is OPEN whether $\sup_\mathcal{F} \tau_\mathrm{mem} < M^2 K^\star$. For rates tied by a symmetry group, Proposition 3 transfers only if I^* is a union of orbits.

For three-phonon events, M has a reading in terms of sub-networks.

Proposition 4 (Sub-network form of M; PROVED)**.** *Let $\boldsymbol{\zeta}_\gamma = \mathsf{W}^{-1}\mathbf{s}_\gamma$ as in Eq. (15), and for $I \in \mathfrak{B}$ let S_I be the matrix with columns $\mathbf{s}_\gamma$, $\gamma \in I$. Then $\ker\mathsf{S}_I^\mathsf{T} = \mathrm{span}\{\boldsymbol{\varepsilon}, \boldsymbol{\phi}_I\}$ for some $\boldsymbol{\phi}_I \notin \mathrm{span}\{\boldsymbol{\varepsilon}\}$, and*

$$\mathbf{y}_I = \frac{\Pi_\mathcal{H}\mathsf{W}\boldsymbol{\phi}_I}{\mathbf{j}^\mathsf{T}\mathsf{W}\boldsymbol{\phi}_I}, \qquad |\mathbf{y}_I| = \frac{|\Pi_\mathcal{H}\mathsf{W}\boldsymbol{\phi}_I|}{|\mathbf{j}^\mathsf{T}\mathsf{W}\boldsymbol{\phi}_I|}. \tag{26}$$

Proof. Since W is invertible, the $\boldsymbol{\zeta}_\gamma$ ($\gamma \in I$) are independent if and only if S_I has full column rank $n - 2$; then $\dim\ker\mathsf{S}_I^\mathsf{T} = 2$, and $\boldsymbol{\varepsilon} \in \ker\mathsf{S}_I^\mathsf{T}$ by energy conservation. The conditions $\boldsymbol{\zeta}_\gamma^\mathsf{T}\mathbf{y} = \mathbf{s}_\gamma^\mathsf{T}\mathsf{W}^{-1}\mathbf{y} = 0$ ($\gamma \in I$) say $\mathsf{W}^{-1}\mathbf{y} \in \ker\mathsf{S}_I^\mathsf{T}$, i.e., $\mathbf{y} = a\,\mathcal{V}^{1/2}\mathbf{e} + b\,\mathsf{W}\boldsymbol{\phi}_I$. The condition $\mathbf{y} \in \mathcal{H}$ gives $\mathbf{y} = b\,\Pi_\mathcal{H}\mathsf{W}\boldsymbol{\phi}_I$, and $\mathbf{j}^\mathsf{T}\mathbf{y} = 1$ with $\mathbf{j} \perp \mathbf{e}$ gives $b = 1/(\mathbf{j}^\mathsf{T}\mathsf{W}\boldsymbol{\phi}_I)$; the denominator is nonzero because $I \in \mathfrak{B}$. □

The vector $\boldsymbol{\phi}_I$ is a conserved quantity of the events in I other than energy, $\mathbf{s}_\gamma^\mathsf{T}\boldsymbol{\phi}_I = 0$ for $\gamma \in I$. M is large when some $(n-2)$-event sub-network has such a hidden invariant with a small overlap with the current. By Proposition 3, large memory requires rates concentrated on such a sub-network, which turns its invariant into a slow, weakly current-coupled relaxation mode, the mechanism of Theorem 1.

Corollary 1 (Width of the feasible range; PROVED)**.** *If* $\mathcal{F}(\mathbf{r}^\star, K^\star)$ *is nonempty, then* $\sup_{\mathcal{F}} \tau_{\mathrm{mem}} \,/\, \inf_{\mathcal{F}} \tau_{\mathrm{mem}} \le M^2|\mathbf{j}|^2$.

Proof. Combine $\tau_{\mathrm{mem}} \le M^2 K^\star$ (Theorem 2) with $\tau_{\mathrm{mem}} \ge K^\star/|\mathbf{j}|^2$ (Proposition 2). □

The structure of the feasible set is convex in part. With the extended value $K_0(g) = \sup_{\mathbf{y}}\{2\mathbf{j}^{\mathsf{T}}\mathbf{y} - \mathbf{y}^{\mathsf{T}}\mathcal{C}(g)\mathbf{y}\}$, equal to $\mathbf{j}^{\mathsf{T}}\mathcal{C}(g)^{+}\mathbf{j}$ if $\mathbf{j} \in \operatorname{range}\mathcal{C}(g)$ and $+\infty$ otherwise, K_0 is convex, lower semicontinuous and nonincreasing in g, with $\partial K_0/\partial g_\gamma = -(\boldsymbol{\zeta}_\gamma^{\mathsf{T}}\mathbf{x})^2$ at admissible g. By the generalized Schur complement,

$$\begin{aligned}&\{g \ge 0 :\ \operatorname{diag}\mathcal{C}(g) = \mathbf{r}^\star,\ K_0(g) \le K^\star\}\\ &= \left\{g \ge 0 :\ \operatorname{diag}\mathcal{C}(g) = \mathbf{r}^\star,\ \begin{pmatrix}\mathcal{C}(g) & \mathbf{j}\\ \mathbf{j}^{\mathsf{T}} & K^\star\end{pmatrix} \succeq 0\right\}, \quad (27)\end{aligned}$$

so the infimum of K_0 at fixed lifetimes is a semidefinite program (PROVED). The set contains nonadmissible rates; the infimum over admissible rates coincides with it whenever the lifetime constraint admits an admissible point. No convexity of τ_{mem} is claimed.

D. What bounds the memory: amplitude information

Proposition 5 (Per-event prior; PROVED)**.** *Let* g^{ref} *be admissible and* $F > 0$*. Every* g *with* $g_\gamma \ge g_\gamma^{\mathrm{ref}}/F$ *for all* $\gamma \in \mathcal{E}$ *is admissible and satisfies*

$$\tau_{\mathrm{mem}}(g) \le \frac{F}{\Gamma_{\min}\big(\mathcal{C}(g^{\mathrm{ref}})\big)}, \quad (28)$$

where $\Gamma_{\min}$ *is the smallest eigenvalue on* $\mathcal{H}$*. No lifetime or dc data enter, and the bound holds with or without symmetry constraints on the rates. Conversely,* $\tau_{\mathrm{mem}}(g) > \tau$ *requires* $\min_\gamma g_\gamma/g_\gamma^{\mathrm{ref}} < [\tau\,\Gamma_{\min}(\mathcal{C}(g^{\mathrm{ref}}))]^{-1}$.

Proof. $\mathcal{C}(g) - F^{-1}\mathcal{C}(g^{\mathrm{ref}}) = \sum_\gamma (g_\gamma - g_\gamma^{\mathrm{ref}}/F)\boldsymbol{\zeta}_\gamma\boldsymbol{\zeta}_\gamma^{\mathsf{T}} \succeq 0$, so $\Gamma_{\min}(\mathcal{C}(g)) \ge \Gamma_{\min}(\mathcal{C}(g^{\mathrm{ref}}))/F > 0$, which implies admissibility, and Lemma 1 gives $\tau_{\mathrm{mem}}(g) \le 1/\Gamma_{\min}(\mathcal{C}(g))$. The converse follows by applying the bound with $F = \max_\gamma g_\gamma^{\mathrm{ref}}/g_\gamma$. □

The information that confines τ_{mem} is therefore microscopic: a statement about every event rate relative to a reference, which is what an anharmonic amplitude model provides and what static data do not. The quantitative consequences for AlN are reported in Sec. V.

E. Infrared criterion

We now let the mesh be refined. Consider uniform Γ-centered meshes with N_1 points per reciprocal-lattice direction in spatial dimension d ($N_{\mathbf{k}} = N_1^d$), the Γ point excluded. With the normalization (4), mode sums $\sum_\nu F(\mathbf{k}_\nu, s_\nu)/\mathcal{V}$ are Riemann sums for $(2\pi)^{-d}\sum_s \int_{\mathrm{BZ}} F\, d^d\mathbf{k}$. Write $j_\nu = \mathcal{V}^{-1/2} J(\mathbf{k}_\nu, s_\nu)$ with $J = w\varepsilon v_i$, and define the RTA moments

$$\mathcal{M}_m = \sum_s \int_{\mathrm{BZ}} J(\mathbf{k}, s)^2\, r(\mathbf{k}, s)^{-m}\, \frac{d^d\mathbf{k}}{(2\pi)^d}, \qquad m = 0, 1, 2, \quad (29)$$

so that $|\mathbf{j}|^2 \to \mathcal{M}_0$, $K_{\mathrm{RTA}} \to \mathcal{M}_1$ and $\tau_{\mathrm{mem}}^{\mathrm{RTA}} = \mathcal{M}_2/\mathcal{M}_1$. On each acoustic branch, with $k = |\mathbf{k}|$ and $\hat{\mathbf{k}} = \mathbf{k}/k$, the hypotheses are:

(H1) $r(\mathbf{k}) = r_0(\hat{\mathbf{k}})\, k^\alpha[1 + o(1)]$ as $k \to 0$, uniformly in $\hat{\mathbf{k}}$, with $0 < r_- \le r_0(\hat{\mathbf{k}}) \le r_+ < \infty$;

(H2) $J(\mathbf{k}) \to J_0(\hat{\mathbf{k}})$ as $k \to 0$, uniformly in $\hat{\mathbf{k}}$; J is bounded on the Brillouin zone; and $0 < \sum_{\mathrm{ac}} \int_{S^{d-1}} J_0^2\, r_0^{-m}\, d\Omega < \infty$ for the orders m considered;

(H3) away from $\mathbf{k} = 0$, and on optical branches, r is bounded below by a positive constant.

For a branch with linear dispersion and sound velocity v_s, $w\varepsilon \to k_B T$ and $J_0^2 = (k_B T v_s)^2 \hat{k}_i^2$, which vanishes on the plane $\hat{k}_i = 0$; (H2) requires only that the angular integral be positive. For the event operator one more hypothesis is needed:

(H4) for the admissible operators $\mathcal{C}_{N_1}$ on the meshes and $\mathbf{x} = \mathcal{C}_{N_1}^{+}\mathbf{j}$, there are $c > 0$ and $k_0 > 0$, independent of N_1, such that $|x_\nu| \ge c\,|j_\nu|/r_\nu$ for every mode with $k_\nu < k_0$.

Theorem 3 (Infrared criterion)**.** *Let the meshes, the moments* $\mathcal{M}_m$ *and hypotheses (H1)–(H4) be as above.*

(i) RTA moments *(*PROVED*). Assume (H1)–(H3). Then* $\mathcal{M}_m < \infty$ *if and only if* $m\alpha < d$*; for* $m\alpha = d$ *the moment restricted to* $k > k_{\min}$ *diverges like* $\log(1/k_{\min})$*. In the RTA the conductivity is therefore finite if and only if* $\alpha < d$*, and the memory moment if and only if* $2\alpha < d$*. If, in addition,* J_0 *and* r_0 *are continuous, then on the meshes* $\tau_{\mathrm{mem}}^{\mathrm{RTA}}$ *converges for* $2\alpha < d$ *and grows like* $\log N_1$ *for* $2\alpha = d$*, like* $N_1^{2\alpha-d}$ *for* $d/2 < \alpha < d$*, and like* $N_1^d/\log N_1$ *for* $\alpha = d$*.*

(ii) Event operator *(*DERIVED UNDER ASSUMPTIONS*, namely (H4)). Suppose that* $\mathbf{r} = \operatorname{diag}\mathcal{C}_{N_1}$ *satisfies (H1)–(H3) uniformly in* N_1*, with* J_0 *and* r_0 *continuous, and that (H4) holds. If* $2\alpha \ge d$*, then* $N = |\mathcal{C}_{N_1}^{+}\mathbf{j}|^2 \to \infty$ *and* $\tau_{\mathrm{mem}} \ge N^{1/2}/|\mathbf{j}| \to \infty$ *as* $N_1 \to \infty$*, with no condition on* K_0*; for* $2\alpha > d$*,* $\tau_{\mathrm{mem}} \gtrsim N_1^{\alpha - d/2}$*, and for* $2\alpha = d$*,* $\tau_{\mathrm{mem}} \gtrsim (\log N_1)^{1/2}$*. These are lower bounds; two-sided mesh exponents for the event operator do not follow.*

(iii) Cone constant *(*DERIVED UNDER ASSUMPTIONS*, namely (H4)). Under the hypotheses of (ii),*

$M_{N_1}^2 K_{0,N_1} \to \infty$, where M_{N_1} is the constant of Theorem 2 for the event family on the mesh; if, in addition, K_{0,N_1} stays bounded, $M_{N_1} \to \infty$.

Proof. (i) By (H3) and the boundedness of J, the integrand of Eq. (29) is bounded outside a ball $k < k_0$. Inside it, (H1)–(H2) give $J^2 r^{-m} = [J_0(\hat{\mathbf{k}})^2 r_0(\hat{\mathbf{k}})^{-m} + \epsilon(\mathbf{k})]\, k^{-m\alpha}$ with $\sup_{\hat{\mathbf{k}}} |\epsilon| \to 0$ as $k \to 0$; the error is additive because J_0 may vanish on a plane. In polar coordinates $d^d\mathbf{k} = k^{d-1} dk\, d\Omega$ the ball contributes $\int_{k_{\min}}^{k_0} k^{d-1-m\alpha}[A + \bar{\epsilon}(k)]\, dk$, where $A > 0$ is the angular integral of (H2) and $\bar{\epsilon}(k) \to 0$. Choosing k_0 with $|\bar{\epsilon}| \le A/2$ bounds this between $A/2$ and $3A/2$ times $\int_{k_{\min}}^{k_0} k^{d-1-m\alpha} dk$, which stays bounded as $k_{\min} \to 0$ if and only if $m\alpha < d$ and equals $\log(k_0/k_{\min})$ for $m\alpha = d$. For the mesh laws, write $\mathbf{k} = \mathsf{B}\mathbf{n}/N_1$ with $\mathbf{n} \in \mathbb{Z}^d$ and B the reciprocal-lattice matrix. Up to a constant factor, the small-k part of the mesh sum is $N_1^{m\alpha-d} \sum[\varphi(\mathbf{n}) + o(1)\, |\mathsf{B}\mathbf{n}|^{-m\alpha}]$ over $0 < |\mathsf{B}\mathbf{n}| \lesssim N_1$, where $\varphi(\mathbf{n}) = (J_0^2 r_0^{-m})(\mathsf{B}\mathbf{n}/|\mathsf{B}\mathbf{n}|)\, |\mathsf{B}\mathbf{n}|^{-m\alpha}$ is continuous, nonnegative, homogeneous of degree $-m\alpha$ and has a positive integral over the shell $1 \le |\mathsf{B}\mathbf{u}| < 2$. On each dyadic shell $2^l \le |\mathsf{B}\mathbf{n}| < 2^{l+1}$ the sum of φ is a Riemann sum, $2^{l(d-m\alpha)}[c_* + o(1)]$ with $c_* > 0$, and the $o(1)$ term is handled as in the integral. Summing over $l \le \log_2 N_1$ gives a sum of order 1, $\log N_1$ or $N_1^{m\alpha-d}$ for $m\alpha < d$, $= d$ or $> d$; for $m\alpha < d$ the shells near the origin contribute uniformly little, so that the Riemann sums converge to $\mathcal{M}_m$. Taking the ratio $\mathcal{M}_2/\mathcal{M}_1$ gives the four cases. (ii) By (H4), $N \ge \sum_{k_\nu < k_0} x_\nu^2 \ge c^2 \sum_{k_\nu < k_0} j_\nu^2 r_\nu^{-2}$, the small-$k$ part of the RTA second moment on the mesh, which diverges for $2\alpha \ge d$ at the rates of part (i). Since $|\mathbf{j}|^2 \to \mathcal{M}_0 < \infty$, Proposition 2 gives $\tau_{\mathrm{mem}} \ge N^{1/2}/|\mathbf{j}| \to \infty$. (iii) Theorem 2 gives $\tau_{\mathrm{mem}} \le M_{N_1}^2 K_{0,N_1}$ on every mesh. □

Example 2 (Finite dc response, infinite slope; PROVED)**.** The Stieltjes function $K(p) = \int_0^1 q^2(q^2 + p)^{-1} dq = 1 - \sqrt{p}\, \arctan(1/\sqrt{p})$ is the RTA response of a one-dimensional branch with $J^2 \propto q^2$ and $r \propto q^2$. It has $K(0) = 1$ and $K(p) = 1 - \frac{\pi}{2}\sqrt{p} + O(p)$: the small-$p$ expansion is not analytic and no Cattaneo time exists. With a cutoff $q > 1/N_1$, $K_0 = 1 - 1/N_1$ and $\tau_{\mathrm{mem}} = N_1$.

Hypothesis (H4) is a property of a given family of operators, not of a class: it states that the long-wavelength part of the solution $\mathbf{x}$ is RTA-like, and it holds with $c = 1$ in the RTA. In three dimensions a finite conductivity requires $\alpha < 3$, whereas a finite memory moment requires $\alpha < 3/2$. The low-frequency three-phonon rates of acoustic phonons follow power laws whose exponents depend on the branch and on the crystal symmetry [56]; whether the operators of a given material satisfy (H1)–(H4), and with which α, is a separate question [**PENDING:** AlN: infrared law of the acoustic rates and status of (H1)–(H4) for the AlN operators, Sec. V]. When the memory moment diverges, a Cattaneo time exists only for a regularized problem: a finite sample, whose boundaries scatter long-wavelength phonons, or a finite frequency band. This motivates the film-regularized memory time of Sec. V.

IV. MODEL OF THE FRONT-FACE MEASUREMENT

A closure enters a thermoreflectance experiment through a model of the film response. This section treats the one-dimensional idealization—plane heating and detection at the front face, no metal transducer, perfect contact—in which the constitutive law of the film enters through two combinations of its parameters. It fixes the model, its depth-scaling gauge and the place of the closure coefficient; the identifiability of the remaining parameters is the subject of a theorem stated in Sec. IV G. The transducer, interface resistances and finite beams of an actual FDTR experiment are treated in Sec. VI.

A. Assumptions

(F1) Heat flows in one dimension across a homogeneous film $0 < z < e$ on a semi-infinite substrate $z > e$; the response is linear and the coefficients are constant.

(F2) The initial state is at rest and there is no interior source. The absorbed flux density $\wp(t)$ enters at $z = 0$, where the temperature rise $\theta(0, t)$ is detected; the front face has no other losses.

(F3) In the film, the energy balance $\rho c\, \partial_t \theta + \partial_z q = 0$ holds together with the one-dimensional form of Eq. (13),

$$\tau_R \partial_t q + q = -\lambda\, \partial_z \theta + L^2 \partial_z^2 q, \qquad L^2 = (1 + \beta)\, \ell^2, \quad (30)$$

obtained for $\mathbf{q} = q(z, t)\, \mathbf{e}_z$, for which $\nabla^2 \mathbf{q} + \beta \nabla(\nabla \cdot \mathbf{q}) = (1 + \beta)\, \partial_z^2 q\, \mathbf{e}_z$. The closure coefficient is $\beta = 2$ for the original GK equation and $\beta = 1/3$ for the conserving closure; $L = 0$ is Cattaneo's law and $\tau_R = L = 0$ Fourier's law.

(F4) Temperature and normal flux are continuous at $z = e$ (perfect contact, with the temperature defined by the local energy density on both sides).

(F5) The substrate obeys Fourier's law with a known effusivity $b_s > 0$.

(F6) The observable is the front-face temperature for all $p = i\omega$, $\omega > 0$, unless a finite set of frequencies is stated.

We write $\theta = T - T_0$, $a = \lambda/\rho c$, $b = (\lambda \rho c)^{1/2}$, $\xi_1 = e/\sqrt{a}$ and

$$\tau_\ell = \frac{L^2}{a} = \frac{(1 + \beta)\, \ell^2}{a}. \quad (31)$$

Three-dimensional models of non-Fourier thermoreflectance impose additional slip and temperature-jump conditions at boundaries [13, 14]; the one-dimensional problem needs none, because after elimination the flux in each layer satisfies a second-order equation (below).

B. Film law and the two combinations

Laplace transformation of the energy balance from the rest state gives $\rho c\, p\, \theta + q' = 0$, hence $q'' = -\rho c\, p\, \theta'$ (primes denote ∂_z). Substituting in Eq. (30), $(1+\tau_R p)\, q = -(\lambda + L^2 \rho c\, p)\, \theta'$, that is $q = -\lambda_{\mathrm{eff}}(p)\, \theta'$ with

$$\lambda_{\mathrm{eff}}(p) = \lambda\, \frac{1+\tau_\ell p}{1+\tau_R p}. \tag{32}$$

Then $\theta'' = -q'/\lambda_{\mathrm{eff}} = \sigma^2 \theta$ with $\sigma^2 = \rho c\, p/\lambda_{\mathrm{eff}}$, and the flux obeys $[\lambda/(\rho c\, p) + L^2]\, q'' = (1+\tau_R p)\, q$, a second-order equation. The film enters the solution only through

$$\sigma e = \xi_1 \Big[\frac{p(1+\tau_R p)}{1+\tau_\ell p}\Big]^{1/2}, \qquad \lambda_{\mathrm{eff}}\, \sigma = b \Big[\frac{p(1+\tau_\ell p)}{1+\tau_R p}\Big]^{1/2}; \tag{33}$$

the two times exchange their roles between the two combinations.

C. Quadrupole and front-face impedance

With the flux counted positive toward $z > 0$, the state at the front face is related to the state at the interface by the thermal quadrupole [**PENDING:** ref: Maillet *et al.*, *Thermal Quadrupoles* (Wiley, 2000); no registered DOI found] [57]

$$\begin{pmatrix} \theta(0) \\ q(0) \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} \theta(e) \\ q(e) \end{pmatrix}, \tag{34}$$

with $A = D = \cosh \sigma e$, $B = \sinh(\sigma e)/(\lambda_{\mathrm{eff}}\sigma)$, $C = \lambda_{\mathrm{eff}}\sigma \sinh \sigma e$ and $AD - BC = 1$. These entries are even functions of σ, so the branch of the square root is immaterial in the film. The semi-infinite substrate has the impedance $Z_s = \theta(e)/q(e) = 1/(b_s\sqrt{p})$ with $\mathrm{Re}\sqrt{p} > 0$, the decaying solution. The front-face impedance $Z = \theta(0)/q(0)$ is

$$Z(p) = \frac{AZ_s + B}{CZ_s + D} = \frac{1}{\lambda_{\mathrm{eff}}\sigma}\, \frac{\varrho(p) + \tanh \sigma e}{1 + \varrho(p) \tanh \sigma e}, \tag{35}$$

where

$$\varrho(p) = \lambda_{\mathrm{eff}}\sigma\, Z_s = \frac{b}{b_s}\Big[\frac{1+\tau_\ell p}{1+\tau_R p}\Big]^{1/2} \tag{36}$$

is a frequency-dependent effusivity ratio; the second form of Eq. (35) is independent of the branch chosen for σ and remains bounded for thick layers, where $\tanh \sigma e$ saturates. The front-face temperature is $\hat\theta(0,p) = Z(p)\, \hat\varphi(p)$, and in the harmonic regime the measured amplitude and phase are $|Z(i\omega)|$ and $\arg Z(i\omega)$. The characteristic frequency $f_c = 1/(2\pi\xi_1^2) = a/(2\pi e^2)$ separates the substrate-dominated regime, $f \ll f_c$, from the film-dominated one.

D. Four combinations and the depth-scaling gauge

Proposition 6 (Dependence and gauge; DERIVED UNDER ASSUMPTIONS)**.** *Assume (F1)–(F6); the following statements are exact within this model.*

(i) $Z(p)$ depends on the film only through $\boldsymbol{\chi} = (\xi_1, b, \tau_R, \tau_\ell)$ and on the substrate only through b_s.

(ii) For every $\mu > 0$ the transformation

$$\begin{aligned} \mathcal{G}_\mu :\quad & e \to \mu e, \quad a \to \mu^2 a, \quad \lambda \to \mu\lambda, \\ & \rho c \to \rho c/\mu, \quad \ell \to \mu \ell, \end{aligned} \tag{37}$$

with τ_R and β fixed, leaves $\boldsymbol{\chi}$, and hence $Z(p)$ for all p, invariant. Infinitesimally, in the variables $(e, a, \lambda, \ell, \tau_R, \beta)$,

$$\big(e\, \partial_e + 2a\, \partial_a + \lambda\, \partial_\lambda + \ell\, \partial_\ell\big)\, Z = 0. \tag{38}$$

(iii) The orbits of $\mathcal{G}_\mu$ are one-dimensional, and fixing any one of e, a, λ, ρc, or ℓ when $\ell > 0$, selects a single point on each orbit.

Proof. (i) follows from Eqs. (33)–(36). (ii) Under $\mathcal{G}_\mu$, $\xi_1 = e/\sqrt{a} \to \mu e/(\mu\sqrt{a})$, $b = (\lambda\rho c)^{1/2} \to (\mu\lambda\, \rho c/\mu)^{1/2}$ and $\tau_\ell = (1+\beta)\ell^2/a \to (1+\beta)\mu^2\ell^2/(\mu^2 a)$; differentiation at $\mu = 1$ gives Eq. (38). (iii) e, a, λ, ρc and ℓ scale as μ, μ^2, μ, μ^{-1} and μ. □

For Fourier films ($\tau_R = \tau_\ell = 0$) this is the non-identifiability of thickness, diffusivity and conductivity shown by Krapez and Rigollet, which holds whatever the frequencies and the noise level [35]. Proposition 6 states that neither a relaxation time nor a nonlocal term removes it, since both enter only through times invariant under depth scaling. The extension is elementary dimensional analysis; we did not find it stated for non-Fourier laws in the literature searched. Independent information that fixes a length scale—a measured thickness, conductivity or heat capacity—selects the point on the orbit; so does an independent velocity v together with the grey kinetic relation $a = v^2\tau_R/3$, conditionally on that relation. Finite beams with known radii and in-plane properties introduce a lateral length; whether this restores the identifiability of (e, a, λ) is part of the pending analysis of Sec. IV G.

Corollary 2 (Closure coefficient and nonlocal length; DERIVED UNDER ASSUMPTIONS (F1)–(F6))**.** *The closure coefficient β and the nonlocal length ℓ enter Z only through $\tau_\ell = (1+\beta)\ell^2/a$. Even with e and a known, all pairs (β, ℓ) on the curve $(1+\beta)\ell^2 = a\tau_\ell$ give the same response. In particular, one-dimensional front-face data do not distinguish the original ($\beta = 2$) from the conserving ($\beta = 1/3$) closure when ℓ is free, and the lengths inferred from one fitted τ_ℓ differ by $\ell_{\beta=1/3}/\ell_{\beta=2} = 3/2$.*

Distinguishing the closures therefore requires a multi-dimensional geometry or independent constitutive information. Thermodynamics does not fix β either: with the

entropy density $s_{\mathrm{eq}} - \tau_R q^2/(2\lambda T_0^2)$ and the entropy flux $q/T + L^2 q\,\partial_z q/(\lambda T_0^2)$, the entropy production to second order in the perturbation is $(q^2 + L^2(\partial_z q)^2)/(\lambda T_0^2) \geq 0$ for every $L^2 \geq 0$, within extended irreversible thermodynamics [58], and nonequilibrium-thermodynamic derivations treat the Laplacian and grad-div coefficients as independent parameters [59].

E. Fourier resonance

If $\tau_R = \tau_\ell$, then $\lambda_{\mathrm{eff}} = \lambda$, $\sigma e = \xi_1\sqrt{p}$ and $\lambda_{\mathrm{eff}}\sigma = b\sqrt{p}$, and Z coincides for all p with the response of a Fourier film with the same (ξ_1, b). This is the Fourier resonance, named by Both *et al.* [41] and analyzed by Kovács [42]; the equivalence concerns the boundary-value problem (F1)–(F6) and does not extend to interior sources or arbitrary initial states. Its position in terms of kinetic times depends on the closure. With the grey coefficients of Sec. II D, $a = v^2\tau_R/3$ and $\ell^2 = v^2\tau_c\tau_R/5$, the conserving closure gives $\tau_\ell = 4\tau_c/5$ and

$$\frac{\tau_\ell}{\tau_R} = \frac{4\,\tau_N}{5\,(\tau_N + \tau_R)} < \frac{4}{5}, \tag{39}$$

so it has no Fourier resonance at positive times, an elementary consequence of the published closure [20]. The original closure gives $\tau_\ell = 9\tau_c/5$ and a resonance at $\tau_R/\tau_N = 4/5$; with τ_c replaced by τ_N it gives $\tau_R/\tau_N = 9/5$. Neither ratio satisfies $\tau_N \ll \tau_R$, which the closure itself assumes, so the resonance line is a closure-dependent formal marker, not a physical boundary between transport regimes.

F. Limits and checks

For $\tau_R = \tau_\ell = 0$, Eq. (35) is the classical quadrupole response of a coating on a substrate, with $\varrho = b/b_s$. For a thick film, $\mathrm{Re}\,\sigma e \to \infty$ and $Z \to 1/(\lambda_{\mathrm{eff}}\sigma)$, the impedance of a half-space, with phase

$$\arg Z(i\omega) = -\frac{\pi}{4} + \frac{1}{2}\big[\arctan(\omega\tau_R) - \arctan(\omega\tau_\ell)\big]. \tag{40}$$

For Cattaneo's law ($\tau_\ell = 0$) the phase rises from $-45°$ to $0°$, the closed form of Camacho de la Rosa *et al.* [15]; Eq. (40) also agrees with the one-dimensional limit of the GK model of Hennessy and Myers [14] after conversion to the time convention $e^{+i\omega t}$. Replacing p by $p(1+\tau_R p)$ throughout the Fourier quadrupole gives the correct argument σe for Cattaneo's law but a flux coefficient that is wrong by the factor $1+\tau_R p$; the half-space phase then runs from $-45°$ toward $-90°$. A Cattaneo film propagates disturbances at the finite speed $(a/\tau_R)^{1/2}$ [9]; for $\ell > 0$ the term $L^2\partial_z^2 q$ makes the principal part parabolic and the propagation speed unbounded.

G. Identifiability of the four combinations

Proposition 6 bounds what the measurement can identify: at most $\boldsymbol{\chi}$. Whether the map $\boldsymbol{\chi} \mapsto Z(\cdot\,;\boldsymbol{\chi})$ is locally and globally injective—structural identifiability in the sense of Ref. [60]—and on which exceptional set it degenerates, is the following statement.

[**PENDING:** Theorem: generic rank four and degenerate set]

V. ALN: FILM-REGULARIZED MEMORY TIME

[**PENDING:** AlN first-principles results: film-regularised memory-time ranges. Planned content: (i) event geometry, reference operator and validation against an independent solver; infrared control of the bulk memory moment on the event operator and in the RTA (mesh series), and the status of hypotheses (H1)–(H4); (ii) the minimum and maximum of $\tau_{\mathrm{mem}}/\tau_{\mathrm{stat}}$ in the event-cone class for the Debye and AlN event sets, with certified witnesses and the constant M; (iii) the film-regularized memory time $\tau_{\mathrm{mem}}(L)$ of the conserving closure in the force-constant-parametrized rate class at 30, 100 and 300 nm, with held-out-mesh checks, the regularization used and its limitations; every number with its numerical-error estimate and status label]

VI. DETECTABILITY IN FREQUENCY-DOMAIN THERMOREFLECTANCE

[**PENDING:** FDTR detectability: phase differences and detectability thresholds of the admissible range of the film-regularized memory time for the Au/AlN/sapphire scenario (500 nm AlN, 10 kHz–200 MHz, phase noise 0.01°–0.1°, amplitude noise 1 %), with the declared nuisance sets (instrument gain, phase and delay; interface resistances; film conductivity and capacity; nonlocal time and transducer properties; beam radii), convergence of the Hankel quadrature, frequency grid and Jacobian steps, and an independent one-dimensional check]

VII. DISCUSSION AND CONCLUSIONS

[**PENDING:** conclusions: to be written after Secs. V and VI and the rank theorem are final; restricted to the evidence, with the limitations of the operator classes, of the regularized memory time and of the one-dimensional measurement model stated explicitly]

DATA AVAILABILITY

[**PENDING:** data and code availability statement: repository, archived version and identifiers]

Appendix A: Notation

Tables I and II list the symbols used in the main text. Vectors in mode space and Cartesian vectors are bold; ^{T} denotes transposition, $^{+}$ the Moore–Penrose pseudoinverse and $\succeq 0$ positive semidefiniteness. The time convention is $e^{+i\omega t}$, with Laplace variable p and $p = i\omega$ in the harmonic regime; fluxes are densities per unit area.

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TABLE I. Kinetic symbols (Secs. II and III).

Symbol	Meaning	Where
$\nu = (\mathbf{k}, s)$, n	mode index, number of modes	(K1)
$N_{\mathbf{k}}$, N_1, $\mathcal{V}$, d	number of wave vectors, points per direction, crystal volume, spatial dimension	(K1), III E
ε_ν, $\mathbf{v}_\nu$, f^0_ν	energy, group velocity, Bose occupation	(K1)
w_ν, W	$[f^0_\nu(1+f^0_\nu)]^{1/2}$, $\mathrm{diag}(w_\nu)$	(K2)
c_ν	mode heat capacity $\varepsilon^2_\nu w^2_\nu/(k_B T^2)$	(5)
$\mathbf{y}$	entropy coordinates	(1)
Ω, $\mathcal{C}$	collision operator acting on δf; the same in entropy coordinates	(2)
$\mathbf{e}$, $\mathcal{H}$, $n_{\mathcal{H}}$, $\Pi_{\mathcal{H}}$	energy invariant, its orthogonal complement, $\dim \mathcal{H} = n-1$, orthogonal projector	(3), (K3)
$\mathbf{j} = \mathbf{j}^{(i)}$, q_i	current vector, heat-flux density	(4)
ρc, Λ_i	volumetric heat capacity, ballistic weight	(5)
$\Phi(t)$, $\kappa(t)$	current autocorrelation function, memory kernel	(7)
$K(p)$, $\lambda(p)$, λ	response function, dynamic conductivity, dc conductivity	(8)
Γ_k, $\mathbf{u}_k$, λ_k	relaxon rates, relaxons, relaxon contributions to λ	(9), (11)
$\mathbf{x}$, K_0, N	$\mathcal{C}^+\mathbf{j}$, $K(0)$, $\lvert\mathbf{x}\rvert^2$	(10)
τ_{mem}, τ_{stat}	memory moment $-K'(0)/K(0)$, static time $K_0/\lvert\mathbf{j}\rvert^2$	(10)
τ_N, τ_R, τ_c	normal, resistive and combined times of the Callaway model	(12), (13)
$\mathbf{P}_i$, Π_{N}	crystal momentum, projector onto $\mathrm{span}\{\mathbf{e}, \mathbf{P}_x, \mathbf{P}_y, \mathbf{P}_z\}$	(12)
β, ℓ, L_{v}	closure coefficient, nonlocal length, scale of spatial variation	(13)
$\mathbf{r}$, τ_ν	mode relaxation rates $\mathrm{diag}\,\mathcal{C}$, lifetimes $1/r_\nu$	Def. 3
$\mathfrak{P}(\mathbf{e})$, $\mathfrak{E}(Z)$	positive-semidefinite class, fixed event cone	Def. 4
$Z = (\boldsymbol{\zeta}_\gamma)_{\gamma\in\mathcal{E}}$	event vectors, event set	(14)
$\mathbf{s}_\gamma$, g_γ	stoichiometric vector, event rate	(15)
$\mathcal{F}(\mathbf{r}^\star, K^\star)$	feasible rates for given static data	(16)
$\mathfrak{B}$, $\mathbf{y}_I$, M	bases, basic solutions, cone constant	Def. 5
$\boldsymbol{\psi}_I$, g_I, ϖ_I	cofactor vector, product of rates, convex weight	(22)
S_I, $\boldsymbol{\phi}_I$	stoichiometric matrix of the events in I, hidden invariant	(26)
K_{RTA}, $\varsigma_{\max}$	RTA dc response, largest number of distinct modes per event	Lemma 2
$\Gamma_{\min}$, g^{ref}, F	smallest eigenvalue on $\mathcal{H}$, reference rates, prior factor	(28)
α, r_0, J, J_0	infrared exponent, angular rate coefficient, current density and its long-wavelength limit	(H1)–(H2)
$\mathcal{M}_m$, k_0, $k_{\min}$	RTA moments, infrared radius, cutoff	(29)
H, $\mathbf{h}_k$, $\mathsf{T}(\eta)$, η	Walsh matrix, its columns, block matrix, family parameter	(18)

TABLE II. Symbols of the front-face model (Sec. IV).

Symbol	Meaning	Where
z, e	depth, film thickness	(F1)
θ, q, $\wp$	temperature rise, flux density (positive toward $z>0$), absorbed flux density	(F2)
λ, ρc, a, b	conductivity, volumetric heat capacity, diffusivity, effusivity of the film	IV A
b_s	effusivity of the substrate	(F5)
ξ_1	$e/\sqrt{a}$	IV A
τ_R, ℓ, β, L^2	relaxation time, nonlocal length, closure coefficient, $(1+\beta)\ell^2$	(30)
τ_ℓ	nonlocal time L^2/a	(31)
λ_{eff}, σ	effective conductivity, propagation constant	(32)
A, B, C, D	quadrupole entries	(34)
Z, Z_s, ϱ	front-face and substrate impedances, frequency-dependent effusivity ratio	(35), (36)
f_c	characteristic frequency $1/(2\pi\xi_1^2)$	IV C
$\boldsymbol{\chi}$	film combinations $(\xi_1, b, \tau_R, \tau_\ell)$	Prop. 6
μ, $\mathcal{G}_\mu$	gauge parameter, depth-scaling transformation	(37)

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